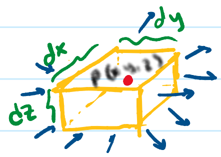
Lecture 1 Planned Notes

Monday, May 19, 2025 10:59 PM

Review

Divergence: $\operatorname{div} \vec{v} = \vec{\nabla}\cdot\vec{v} = \frac{\partial}{\partial x} v_1 + \frac{\partial}{\partial y} v_2 + \frac{\partial}{\partial z} v_3$ where $\vec{v}(x,y,z) = \begin{bmatrix} v_1(x,y,z) \\ v_2(x,y,z) \\ v_3(x,y,z) \end{bmatrix}$

Roughly speaking:



$\vec{\nabla}\cdot\vec{v}\big|_P$ measures the net flow, i.e. outflow − inflow of vector field $\vec{v}$ in a small $dx\,dy\,dz$ box centered at point $P(x,y,z)$

Properties of Divergence

(a) $\operatorname{div}(k\mathbf{v}) = k \operatorname{div} \mathbf{v}$ (k constant)

(b) $\operatorname{div}(f\mathbf{v}) = f \operatorname{div} \mathbf{v} + \mathbf{v} \cdot \nabla f$

(c) $\operatorname{div}(f\nabla g) = f\nabla^2 g + \nabla f \cdot \nabla g$

(d) $\operatorname{div}(f\nabla g) - \operatorname{div}(g\nabla f) = f\nabla^2 g - g\nabla^2 f$

The Laplacian: $\nabla^2 f = \vec{\nabla}\cdot(\vec{\nabla} f) = \frac{\partial^2}{\partial x^2} f + \frac{\partial^2}{\partial y^2} f + \frac{\partial^2}{\partial z^2} f$